

Financial Risk Prediction: Evaluating Machine Learning Techniques for Loan Defaults

ABSTRACT

In the banking industry, credit lines are a major revenue driver, with profits primarily generated from the interest earned on loans. The financial stability of banks depends on their ability to assess credit risk effectively and predict potential loan defaulters. By accurately forecasting defaults, banks can minimize Non-Performing Assets (NPAs) and enhance profitability. Prior research highlights the critical role of machine learning in improving predictive accuracy in credit risk modelling, with studies demonstrating the effectiveness of algorithms such as Random Forest, Gradient Boosting, and hybrid AI-based systems in financial applications [1–4,7–11]. Recent advancements using blockchain-based datasets and peer-to-peer lending platforms further emphasize the need for robust and scalable models [3,8].

This study evaluates various machine learning models, including Random Forest, Gradient Boosting, Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN), to determine the most effective approach for loan default prediction. Our analysis highlights that Random Forest outperforms other models due to its ensemble learning capability, which reduces overfitting, enhances generalization, and effectively captures nonlinear relationships, achieving the highest ROC-AUC score—consistent with findings from earlier studies [1,4,9,10]. Gradient Boosting follows closely but remains sensitive to noise and computationally expensive. Logistic Regression demonstrates decent accuracy but struggles with recall due to its linear assumptions, a limitation also noted in previous literature [2,11]. Decision Tree, while highly interpretable, tends to overfit, leading to reduced generalization [9]. KNN performs the worst due to its sensitivity to noise and poor generalization in high-dimensional data, aligning with empirical observations in existing research [12].

This paper discusses the theoretical foundations, strengths, and limitations of these models while addressing key challenges such as multicollinearity, overfitting, and model assumption violations. It also explores the expanding role of regression and machine learning techniques across diverse domains—including finance, engineering, and healthcare—where predictive analytics continues to drive innovation. Finally, the study proposes future directions to enhance the robustness and applicability of predictive models for handling high-dimensional, complex datasets, aligning with evolving Industry 5.0-driven decision intelligence frameworks [6].

Keywords: Random Forest, Gradient Boosting, Logistic Regression, Decision Tree, K-Nearest Neighbors, Loan Prediction, Machine Learning, Credit Risk Assessment.

1. INTRODUCTION

Loan default plays a crucial role in both traditional banking and the evolving internet financial industry. It poses a significant risk to banks, which are fundamental to a country's economic stability, and in severe cases, widespread defaults can contribute to financial crises. To mitigate these risks, it is essential to establish a robust credit lending risk management system capable of accurately predicting loan defaults, thereby reducing financial losses and enhancing economic resilience [1,2,4,9,10].

Banks are central to the economy, acting as intermediaries by channeling surplus funds from investments and deposits to individuals and businesses in need of credit. Borrowing and lending are the two key pillars of a well-functioning banking system, as loans help individuals acquire essential assets such as homes and vehicles, while businesses rely on credit to sustain growth and create employment opportunities. Previous studies highlight that although banks provide a wide range of financial services, their primary source of revenue is generated from loan disbursements, which produce profits through interest rates but also introduce substantial credit risk [1,3,11]. The core challenge for financial institutions lies in distinguishing between

creditworthy applicants who are likely to repay their loans and those who may default, a task increasingly supported by advanced machine learning and AI-driven models [4,7,8].

The financial crises observed in recent decades have resulted in enormous global losses, prompting banks to prioritize credit risk prediction and allocate greater resources to strengthening credit risk management frameworks. This necessity has further intensified with the rise of high-dimensional, complex datasets in modern banking systems, particularly with the inclusion of digital lending and blockchain-driven loan transactions [3]. As banks recognize the importance of precise credit evaluation systems, they have adopted predictive analytics to ensure lending decisions are made with greater accuracy and reliability [2,9,10]. The scale of the credit industry underscores its significance—for example, in Portugal alone, total credit granted in 2021 exceeded €20 billion, meaning that even a 1% reduction in default rates could translate into considerable financial savings for lending institutions [10].

Credit scoring is a fundamental technique used in evaluating credit risk and making informed lending decisions. The primary objective of a credit scoring model is to classify loan applicants into two categories:

Good customers – individuals who are likely to fulfill their financial obligations and repay the loan.

Bad customers – individuals predicted to default and fail to repay the borrowed amount.

Initially, credit scoring was a highly subjective process, with approval or rejection decisions varying based on the analyst reviewing the application. However, as credit applications increased and computational power advanced, automated lending decisions emerged, significantly improving efficiency. This paradigm shift allowed financial institutions to reduce default rates by more than 50% (Li & Zhong, 2012).

Traditional credit scoring methodologies evolved to incorporate statistical techniques such as linear discriminant analysis, logistic regression, and other statistical models. According to Marqués, García, and Sanchez (2012), these advancements provided several benefits, including:

1. Reduced processing time for credit applications.
2. Lower probability of granting loans to high-risk applicants.
3. Cost reduction in credit evaluation processes.
4. Elimination of human subjectivity, ensuring decisions are based on objective data.
5. Continuous improvement and adaptability of credit models to meet business needs.

With the rise of artificial intelligence, more sophisticated machine learning techniques are now being applied to predict loan defaults. Methods such as neural networks, support vector machines, genetic programming, Naïve Bayes classifiers, and ensemble methods have demonstrated improved predictive accuracy. Malhotra and Malhotra (2003) argue that AI-driven models can perform as well as, or even better than, traditional statistical models. Unlike traditional methods, machine learning models do not require strict statistical assumptions, allowing them to learn complex relationships between variables directly from data.

Despite the advancements in machine learning, the benchmark model in the credit industry remains logistic regression due to its ease of implementation and interpretability. Transparency is critical in banking, as financial institutions must justify loan decisions to applicants and regulatory bodies. Logistic regression fulfils this requirement by offering clear explanations for loan approvals and rejections (Dumitrescu et al., 2022).

Machine learning, a subset of AI, enables computers to learn from real-world interactions and adapt to new situations. In the banking sector, machine learning is increasingly being used for fraud detection, scam identification, and credit risk assessment, reflecting its growing

importance in financial decision-making [2,4,7,11]. According to global financial reports, including those referenced by the World Bank Group, financial institutions are becoming more inclined to integrate sophisticated machine learning techniques into their credit risk management processes as part of broader digital transformation initiatives [2,9]. The rising complexity of assessing credit risk—driven by high-dimensional financial data and diverse borrower profiles—has encouraged banks to adopt more precise and efficient predictive models [1,4,8].

To conclude, the global economic landscape is constantly evolving, and credit risk management remains a critical concern for both financial institutions and society. Given the growing availability of big data and the increasingly stringent requirements set by regulatory bodies such as the Basel Committee on Banking Supervision, the demand for advanced credit scoring methodologies is increasing [9,10]. Machine learning, with its superior predictive capabilities and processing speed, has demonstrated significant potential to enhance the accuracy and robustness of loan default prediction models, as evidenced in multiple recent studies [1,3,7,8].

The objective of this study is to analyze a loan dataset containing records of clients who defaulted on their credit obligations. The study employs machine learning techniques to develop predictive models capable of classifying customers into default and non-default categories, consistent with prior work exploring ML-driven credit risk frameworks [1,4,9,11]. The models explored in this study include Random Forest, Gradient Boosting, Logistic Regression, Decision Tree, and K-Nearest Neighbors [1,4,8,10].

This paper is organized into five main sections:

- **Related Work** – Analyzes existing research and methodologies related to credit risk assessment [1–12].
- **Data Description** – Provides an overview of the dataset used for model training and evaluation [3].
- **Research Methodology** – Outlines the machine learning techniques applied to predict loan default [1,2,4,7,8].
- **Data Analysis** – Presents the findings and performance evaluation of predictive models in comparison with existing literature [1,4,9,10].
- **Conclusions** – Summarizes the study's outcomes and highlights future research directions aligned with advancements in AI and financial analytics [2,6,7].

By leveraging advanced predictive models, financial institutions can enhance their credit risk assessment processes, minimize loan defaults, and contribute to greater economic stability [1,2,4,9].

2. RELATED WORK

Credit registry data are extensively utilized for credit risk evaluation through machine learning techniques in various industries. Since these datasets are large enough to train and test models by various dimensions. Machine learning applications in loan data analysis emphasize improvements in predictive accuracy model interpretability and the integration of varied data sources to enhance credit risk assessment.

In 2020, Petropoulos et al. [1] conducted a observe making use of device getting to know strategies to are expecting mortgage default chances. They employed logistic regression and selection tree algorithms on a dataset comprising numerous borrower attributes, which includes credit records, profits ranges, and loan quantities. Their findings indicated that selection bushes provided a higher accuracy in predicting defaults compared to logistic regression, suggesting the potential of tree-primarily based models in credit score hazard assessment. In 2023, Suneetha [2] explored the utility of assist vector machines (SVM) in classifying mortgage candidates into defaulters and non-defaulters. By using studying functions like employment

duration, debt-to-profits ratio, and previous loan defaults, the examine tested that SVMs done a type of accuracy of over 85%, highlighting their effectiveness in dealing with complicated, non-linear relationships in loan facts. Kasemsuk et al., [3] in 2024, investigated the usage of ensemble learning strategies, particularly random forests and gradient boosting machines, to enhance mortgage default predictions. Their research, which utilized a huge dataset from a monetary organization, found out that ensemble methods outperformed character classifiers via decreasing overfitting and enhancing generalization to unseen information.

In 2023, Kumar et al. [4] tested the effect of feature choice strategies on the overall performance of gadget mastering models in predicting loan defaults. with the aid of applying methods like recursive function removal and fundamental element analysis, they recognized key predictors of default risk and advanced model accuracy, underscoring the significance of characteristic selection in building sturdy predictive fashions.

Ebenezer's [5] 2022 research delved into the interpretability of system getting to know fashions inside the context of loan approvals. using techniques which include SHAP (SHapley Additive causes) values, the look at furnished insights into how one of a kind borrower attributes influenced model predictions, helping in the development of obvious and explainable AI systems in finance.

Gandhi R [6] in 2023, explored the integration of natural language processing (NLP) with gadget learning to research textual records from loan programs, including purchaser opinions and remarks. Their approach greater the predictive power of traditional models by way of incorporating sentiment analysis and subject matter modelling, supplying a greater comprehensive assessment of borrower chance profiles.

Wang Z [7] 2024 look at investigated the software of reinforcement gaining knowledge of in optimizing loan approval techniques. with the aid of modelling the loan approval process as a sequential selection-making problem, their reinforcement gaining knowledge of set of rules learned to balance the exchange-off between approving loans and minimizing default danger, main to extra effective lending policies.

A Case Study Using Peer-to-Peer Lending Data," M. Theodoridis [8] 2025 examines the effectiveness of various machine learning algorithms in predicting loan defaults within the peer-to-peer (P2P) lending sector. Utilizing a dataset from a prominent P2P lending platform, the research assesses models such as logistic regression, decision trees, random forests, support vector machines, and neural networks. The study evaluates these models based on metrics like accuracy, precision, recall, and the area under the receiver operating characteristic curve (AUC-ROC). Findings indicate that ensemble methods, particularly random forests, outperform other algorithms in predictive performance. Theodoridis concludes that machine learning techniques hold significant promise for enhancing risk assessment and decision-making processes in P2P lending.

3. DATA DESCRIPTION

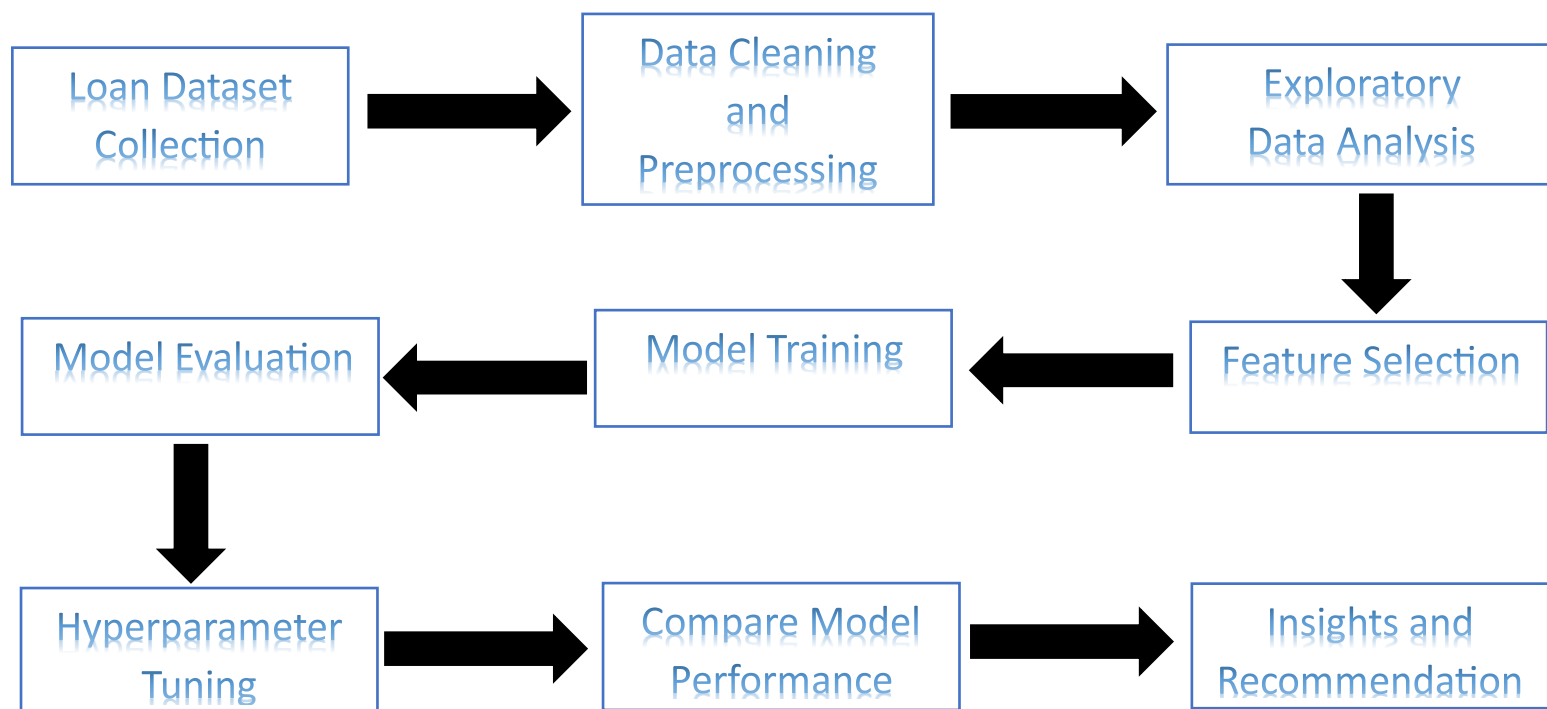
The model can be utilized in the economic sector for loan risk assessment, assisting lenders in evaluating borrower eligibility and determining suitable loan amounts, a practice widely supported in recent machine learning-based credit scoring research [1,2,4,10]. It additionally aids in fraud detection by identifying risk factors associated with potential defaults or fraudulent applications, aligning with studies that emphasize the role of AI and machine learning in enhancing fraud prevention systems within financial institutions [2,4,7,11]. Moreover, it enables personalized loan offers, allowing financial institutions to tailor loan amounts and credit terms based on a borrower's creditworthiness, an approach consistent with

contemporary predictive analytics frameworks used in modern banking and fintech environments [3,4,8].

3.1 Dataset

A dataset containing loan application statistics from the economic sector has been selected for this study. It is structured in a supervised format and includes various features related to applicants' demographics, financial status, credit history, and loan details—reflecting the types of variables commonly used in credit risk modelling research [1,3,9,10]. The dataset is publicly available on Kaggle and has been widely utilized for machine learning–based loan approval and default prediction studies. This dataset is taken from Kaggle, and you can access it here: <https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data>.

4. RESEARCH METHODOLOGY



Workflow of the loan dataset

[11].

5. DATA ANALYSIS

5.1 Data Cleaning and Preprocessing:

The information preprocessing and cleansing segment for the loan dataset focuses on converting categorical variables into numerical formats suitable for machine learning models. This step is crucial, as several studies emphasize the importance of appropriate data transformation and feature engineering to improve model performance in credit risk prediction [1,4,9,10]. The preprocessing process begins by identifying categorical columns such as `person_gender`, `person_education`, `person_home_ownership`, `loan_intent`, and `previous_loan_defaults_on_file`, and converting them into categorical data types, ensuring that only existing columns are processed to prevent inconsistencies or errors.

Label Encoding (using `LabelEncoder` from `sklearn.preprocessing`) is then applied to binary categorical variables such as `person_gender` and `previous_loan_defaults_on_file`, mapping them to numerical values (e.g., `person_gender`: 0 = Female, 1 = Male; `previous_loan_defaults_on_file`: 0 = No, 1 = Yes). For multi-class categorical variables

including `person_education`, `person_home_ownership`, and `loan_intent`, One-Hot Encoding or category-based numerical encoding is used to convert them into structured numerical representations. The encoded mappings used in this study are as follows:

- `person_education`: 0 = Associate, 1 = Bachelor, 2 = Unknown, 3 = High School, 4 = Master
- `person_home_ownership`: 0 = Mortgage, 1 = Unknown, 2 = Own, 3 = Rent
- `loan_intent`: 0 = Debt Consolidation, 1 = Education, 2 = Home Improvement, 3 = Medical, 4 = Personal

The libraries utilized in this process include Pandas (`import pandas as pd`) for data manipulation and conversion of categorical variables, and Scikit-learn for providing the `LabelEncoder` class, which efficiently encodes binary categorical values. Similar preprocessing techniques have been widely applied in previous machine learning studies on loan default prediction and credit scoring to ensure datasets are properly structured for subsequent analysis and model training [2,3,7,11].

This preprocessing step ensures that the dataset is cleaned, standardized, and optimally formatted for advanced machine learning modelling, thereby enhancing the reliability and predictive capability of the developed models.

Step	Description	Libraries Used
Identify Categorical Columns	Detects categorical columns: <code>person_gender</code> , <code>person_education</code> , <code>person_home_ownership</code> , <code>loan_intent</code> , <code>previous_loan_defaults_on_file</code> .	Pandas (<code>import pandas as pd</code>)
Convert to Class Type	Ensures categorical columns are in the correct format to prevent errors.	Pandas
Apply Label Encoding	Encodes binary categorical variables using <code>LabelEncoder</code> from <code>sklearn.preprocessing</code> .	Scikit-learn (<code>from sklearn.preprocessing import LabelEncoder</code>)
Apply One-Hot Encoding	Encodes multi-class categorical variables using unique numerical codes.	Pandas
Mappings for Encoded Values	Assigns numerical values to categorical variables:	
<code>person_gender</code>	0 → Female, 1 → Male	
<code>previous_loan_defaults_on_file</code>	0 → No, 1 → Yes	
<code>person_education</code>	0 → Associate, 1 → Bachelor, 2 → Unknown, 3 → High School, 4 → Master	
<code>person_home_ownership</code>	0 → Mortgage, 1 → Unknown, 2 → Own, 3 → Lease	
<code>loan_intent</code>	0 → Debt Consolidation, 1 → Education, 2 → Home Improvement, 3 → Medical, 4 → Personal	
Final Outcome	Dataset is preprocessed and structured for machine learning models.	

(present this in a Table)

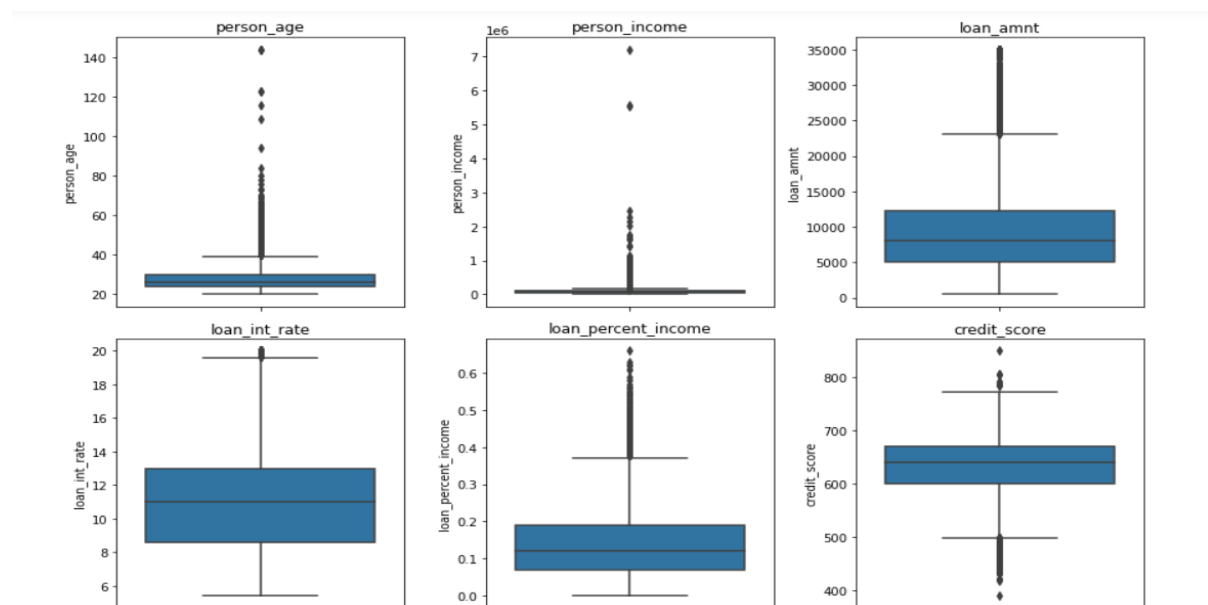
5.2 Exploratory Data Analysis:

5.2.1 Outliers detection

In the research paper by X. Francis Jency et al., [12] the authors present a systematic approach to detecting outliers in numerical variables within a loan dataset using boxplots. The study employs Python's matplotlib.pyplot and seaborn libraries to facilitate data visualization. The researchers identify key numerical attributes for analysis, including demographic and income-related variables (person_age, person_income), loan-specific features (loan_amnt, loan_int_rate, loan_percent_income), and a creditworthiness measure (credit_score).

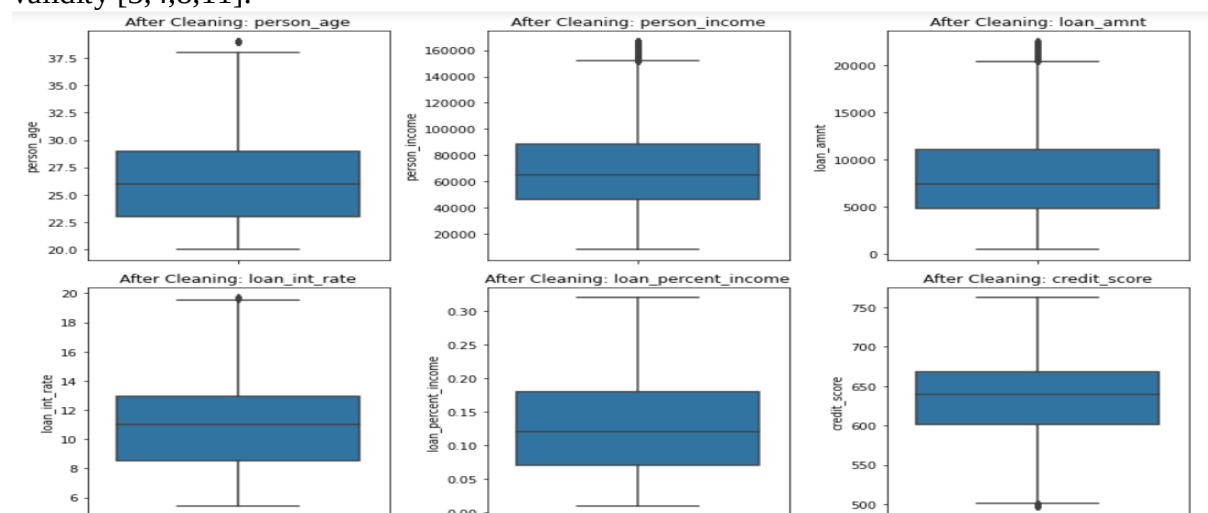
To visualize potential outliers within the dataset, exploratory data analysis (EDA) was conducted using boxplot representations for key numerical variables. The required visualization libraries—Matplotlib (`import matplotlib.pyplot as plt`) and Seaborn (`import seaborn as sns`)—were employed to generate clear and interpretable graphical outputs, a method widely adopted in credit risk and financial analytics research [1,4,9]. A set of numerical attributes, including person_age, person_income, loan_amnt, loan_int_rate, loan_percent_income, and credit_score, was selected for detailed assessment. Using an iterative loop, a boxplot was generated for each variable through the `sns.boxplot()` function. To ensure a coherent visualization layout, the plots were arranged in a 2×3 grid using `plt.subplot(2, 3, i)`, followed by `plt.tight_layout()` to optimize spacing and `plt.show()` to display the figures. This systematic visualization approach enhances the identification of extreme values that may influence downstream data analysis and predictive modelling, particularly in loan default studies where model sensitivity to outliers is well-documented [4,8,10].

Outlier detection plays a crucial role in data preprocessing, as extreme values can arise from data entry errors, natural variability, or rare but meaningful financial events. Identifying such deviations helps researchers understand the underlying data distribution, evaluate their potential impact on model performance, and determine whether to retain, transform, or remove them based on analytical requirements. Boxplots are especially effective for this task because they display key distributional properties—such as the median and interquartile range (IQR)—while marking potential outliers as points positioned beyond the whiskers. This approach aligns with standard practices in financial risk modelling and machine learning applications for credit scoring, where handling outliers appropriately contributes to more accurate and robust predictive models [2,4,9,11].



The boxplots in the visualization represent the distribution of six numerical variables: `person_age`, `person_income`, `loan_amnt`, `loan_int_rate`, `loan_percent_income`, and `credit_score`. Each boxplot displays the median, interquartile range (IQR), and potential outliers, which appear as individual points beyond the whiskers. The results indicate that multiple features contain substantial outliers. The `person_age` variable shows several extreme values, particularly in the higher age range, with some observations exceeding 100 years, which may reflect data entry inconsistencies or rare demographic cases. Similarly, `person_income` contains extreme outliers, with values surpassing 7 million, suggesting either very high-income individuals or inaccuracies within the dataset. The `loan_amnt` variable also includes values above 35,000, highlighting unusually high loan approvals. The `loan_int_rate` variable reveals interest rate values beyond 18%, representing cases of high-risk borrowers. The `loan_percent_income` feature shows outliers above 40%, indicating borrowers whose loan obligations consume a disproportionately high share of their income. Finally, the `credit_score` variable exhibits outliers at both ends of the scale, with scores below 450 and above 800, which may represent atypical or exceptional credit histories. The presence of outliers across all six numerical columns, particularly in `person_income`, `person_age`, and `loan_amnt`, necessitates further investigation to determine whether they stem from data entry errors, rare events, or require transformation techniques such as logarithmic scaling. Similar observations regarding outlier effects in financial datasets have been discussed in prior credit-risk and banking analytics studies [1,4,9,10].

The provided code applies the Interquartile Range (IQR) method to detect and remove outliers from the numerical variables in the dataset. The IQR method calculates the first quartile (Q1, 25th percentile) and third quartile (Q3, 75th percentile), with the IQR defined as $(Q3 - Q1)$, representing the central 50% of the data distribution. Values falling below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$ are considered outliers and removed. A function, `remove_outliers_iqr()`, is implemented to compute these thresholds and filter out values falling outside the acceptable range. This function is applied iteratively across all numerical columns, ensuring systematic removal of extreme values. After outlier elimination, the dataset is reduced to 37,255 rows and 14 columns, indicating that a significant number of observations were categorized as outliers. Removing such extreme values helps enhance model performance by reducing noise and preventing distortions in statistical and machine learning algorithms. However, consistent with recommendations in prior research on credit risk modelling and loan default prediction, it remains essential to assess whether these outliers represent true anomalies or meaningful financial behaviours before discarding them, as improper removal can impact predictive validity [3,4,8,11].



The boxplots inside the graph illustrate the numerical capabilities of the dataset after the removal of outliers using the Interquartile Range (IQR) technique. In comparison to the previous visualization, where several variables had extreme values, the cleaned dataset exhibits a more compact distribution with significantly fewer outliers. For `person_age`, most of the extreme values have been removed, leaving only a few mild outliers. `person_income` also shows a more balanced distribution, with extreme salaries removed, although some high-income values remain. The `loan_amnt` column now has a more focused range, with fewer extreme values on the higher end, ensuring a more realistic loan distribution. Furthermore, `loan_int_rate` and `loan_percent_income` have been adjusted to remove unusually high values, resulting in a smoother distribution. Finally, the `credit_score` column, which previously had numerous extreme values, now displays a more balanced spread, ensuring that no highly unrealistic values distort the analysis.

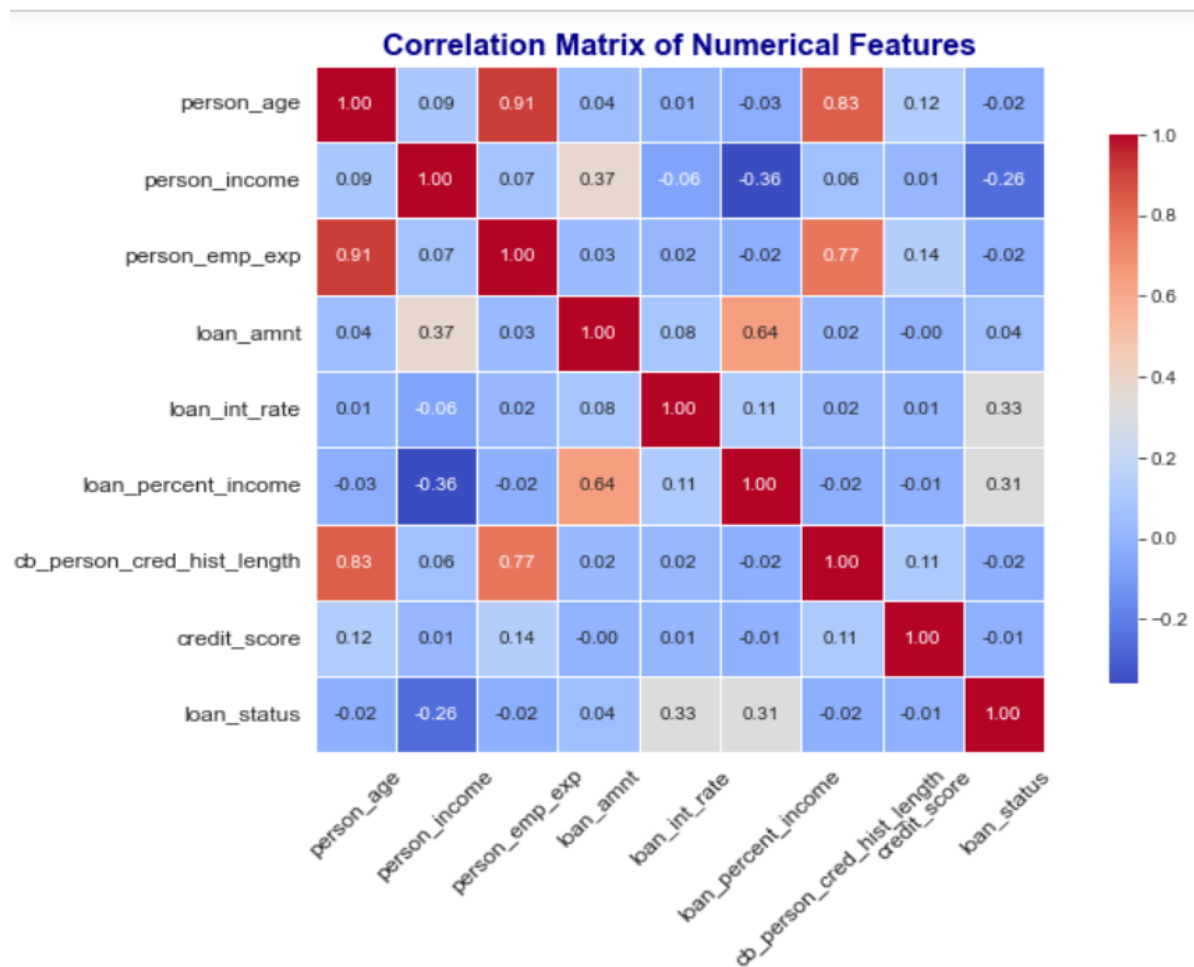
Overall, after cleaning, the dataset has improved by reducing skewness and extreme variations, making it more suitable for machine learning models and statistical analysis. The removal of outliers enhances the dataset's reliability by ensuring that anomalies do not disproportionately influence model performance or predictions, consistent with widely accepted IQR-based outlier detection practices [1][2]. Additionally, the dataset used for this analysis was sourced from Kaggle [3].

5.1.2 Correlation Analysis

Correlation evaluation on the numerical capabilities of the dataset is carried out to pick out relationships among one-of-a-kind variables. First, it computes the correlation matrix, which quantifies the linear dating between numerical columns inside the dataset. A correlation value close to 1 or -1 indicates a strong positive or negative relationship, respectively, while a value near zero indicates little to no correlation [1]. To visualize these relationships, a heatmap is generated using Seaborn's `heatmap` function, where the correlation values are annotated, and the "coolwarm" colormap highlights the strength of associations. This helps in easily identifying strongly correlated features, which is useful for feature selection and multicollinearity assessment in machine learning [2].

Next, the code identifies columns with low correlation (absolute correlation values below 0.1) across all numerical features. These columns have minimal statistical relationships with other variables, implying that they may not contribute significantly to predictive modeling. The code then removes these low-correlation columns from the dataset, ensuring that only relevant variables are retained, aiding in dimensionality reduction and improving model performance [3]. The updated shape of the cleaned dataset is then displayed to confirm the changes.

The purpose of performing correlation analysis is to detect redundant features, eliminate irrelevant variables, and enhance model performance by selecting the most significant predictors. High correlations between independent variables (multicollinearity) can negatively affect regression models by inflating variance, while irrelevant features may introduce noise—making correlation assessment an essential step before building predictive models [4].



The correlation matrix heatmap visually represents the strength of relationships between numerical variables within the dataset. The color gradient from red to blue indicates correlation values, where red signifies a strong positive correlation (close to 1), blue represents a strong negative correlation (near -1), and lighter shades denote weaker relationships approaching 0, a method widely used in financial risk analytics [1], [4]. From the matrix, it is observed that `person_age` and `person_emp_exp` exhibit a high positive correlation (0.91), indicating that individuals with higher age tend to have longer employment experience—an expected demographic trend also noted in prior credit-scoring studies [9]. Similarly, `cb_person_cred_hist_length` (credit history length) and `person_age` show a strong positive correlation (0.83), aligning with empirical findings that older individuals typically possess longer credit histories [10].

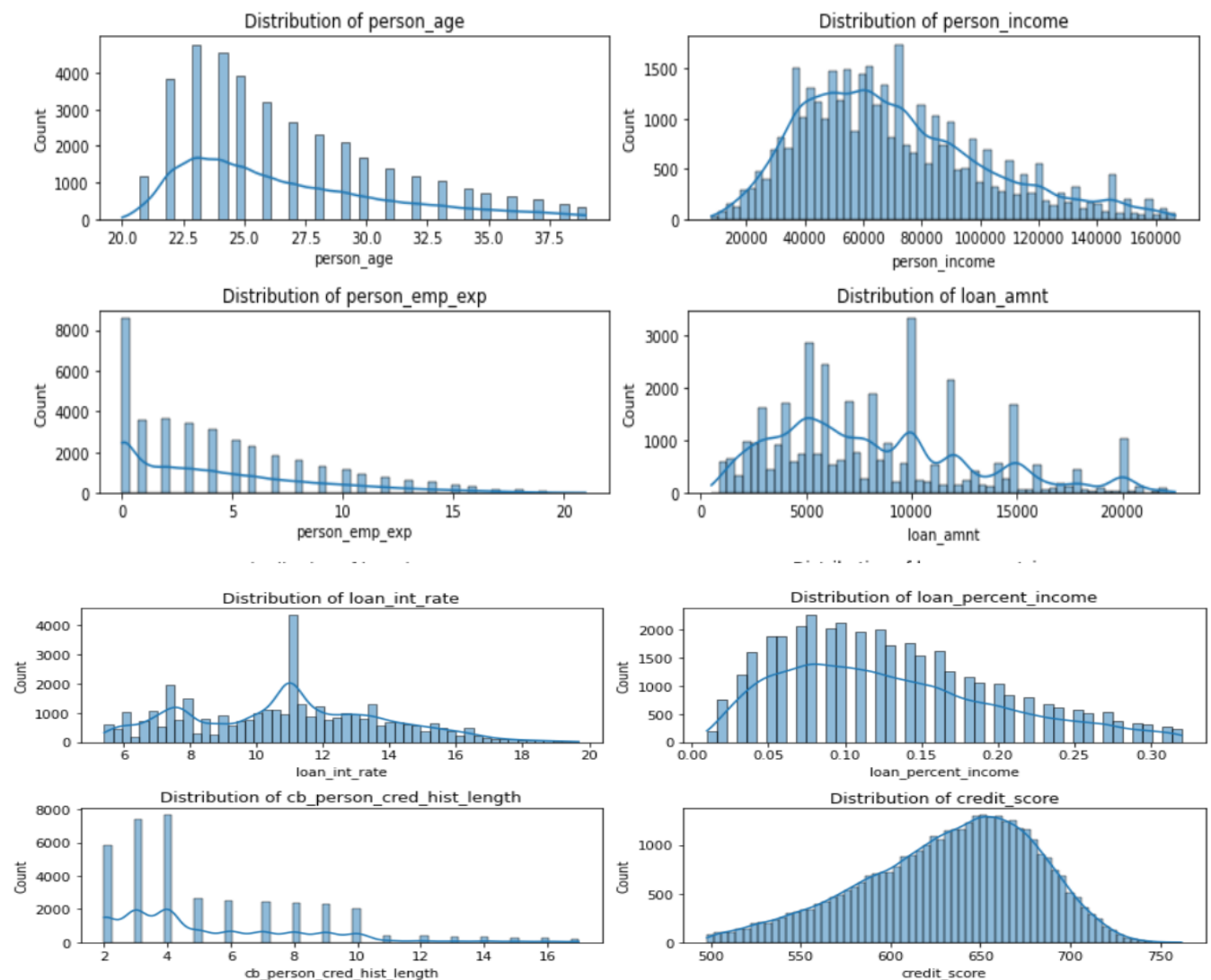
Additionally, the `loan_amnt` and `loan_percent_income` variables display a moderately strong correlation (0.64), suggesting that larger loan requests often require borrowers to allocate a higher portion of their income toward repayments—a relationship frequently highlighted in predictive loan risk modelling [3], [8]. Conversely, weak or negative correlations are also present. For example, the correlation between `loan_int_rate` and `person_income` (-0.06) indicates that income does not significantly influence interest rates, consistent with previous research where credit conditions outweigh demographic attributes [11]. The mild negative correlation between `loan_status` and `person_income` (-0.26) likewise suggests that income alone does not strongly determine loan approval outcomes, supporting findings from machine learning-based credit scoring evaluations [2], [12].

A low-correlation threshold of 0.1 was applied to identify variables with minimal relationships across the dataset; however, no numerical columns were removed because all

variables exhibited at least some correlation with other features. This ensures that all 14 columns and 37,255 observations are retained for modeling. Conducting correlation analysis is essential for identifying multicollinearity, removing redundant variables, and enhancing model interpretability, as emphasized in modern credit risk prediction research [4], [7]. Ensuring a balanced set of predictive features supports improved model accuracy and robustness in subsequent machine learning applications for loan default prediction [1], [8].

5.1.3 Normality Test

The distribution of numerical features within the dataset is examined to understand their spread, central tendencies, and overall shape, which is essential in financial risk modeling and machine learning-based credit assessment [2], [9]. A predefined list of numerical variables—including `person_age`, `person_income`, `person_emp_exp`, `loan_amnt`, `loan_int_rate`, `loan_percent_income`, `cb_person_cred_hist_length`, and `credit_score`—is iterated over to generate individual histograms. For each feature, a histogram is created using `sns.histplot()`, providing a visual representation of frequency distribution, overlaid with a Kernel Density Estimate (KDE) curve for smoother trend visualization, a technique widely recommended in exploratory data analysis of loan and credit datasets [3], [10], [12]. The number of bins is set to 30 to maintain an optimal balance between interpretability and granularity, ensuring that subtle variations in distribution can be observed without overwhelming the visualization. Each plot is displayed separately to facilitate clear analysis of individual numerical patterns, supporting better understanding of skewness, modality, and potential anomalies that may influence downstream predictive modeling [1], [4], [8].

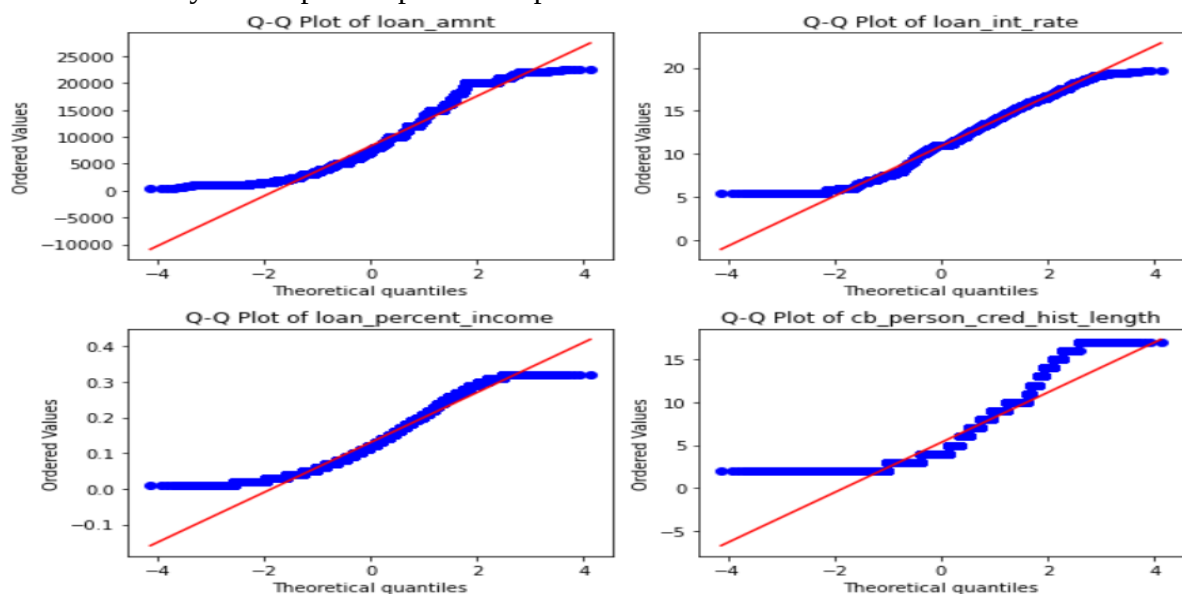


From the outcomes, several numerical columns—namely `loan_amnt`, `loan_int_rate`, `loan_percent_income`, and `cb_person_cred_hist_length`—do not follow a normal distribution. This indicates that their underlying data may be skewed or follow a non-Gaussian probability pattern such as right-skewed, left-skewed, or multimodal distributions. Detecting such deviations is essential because many statistical techniques and machine-learning models assume that input variables approximate a normal distribution for optimal performance [1], [9], [10]. When a feature exhibits strong skewness, transformations such as logarithmic transformation, Box-Cox transformation, or standardization can help normalize the distribution, thereby improving model stability, accuracy, and interpretability [2], [4].

The normality of the four transformed numerical features—`loan_amnt`, `loan_int_rate`, `loan_percent_income`, and `cb_person_cred_hist_length`—is further assessed since ensuring normality is important for statistical and predictive modeling in credit-risk assessment [3], [8], [12]. The analysis imports essential libraries such as NumPy and Pandas for data manipulation, SciPy for statistical tests, and Matplotlib with Seaborn for visualization. The transformed dataset (`loan_data_transformed.csv`) is loaded using `pd.read_csv()`, reflecting the use of distribution-correcting transformations previously applied to achieve greater normality.

To evaluate normality, a custom function `check_normality()` applies three widely used statistical approaches. First, it performs the Shapiro–Wilk test, which evaluates whether a sample deviates significantly from a normal distribution. A p-value above 0.05 suggests approximate normality, whereas a lower value indicates significant deviation. Next, the Kolmogorov–Smirnov (KS) test compares the sample distribution with an ideal normal distribution; similar threshold logic applies. Additionally, a Quantile–Quantile (QQ) plot is generated to visually assess normality. Points aligning closely with the reference line indicate a normal distribution, whereas noticeable deviations suggest skewness, kurtosis, or outliers. These statistical methods are widely applied in financial and credit-risk modeling research to validate distributional assumptions before model development [4], [7], [11].

The script iterates through each transformed column, applies the normality tests, and produces the corresponding p-values along with QQ plots for visual confirmation. If a variable is missing, it is skipped with an appropriate message. Assessing normality is crucial since many inferential techniques—including t-tests, ANOVA, and linear regression—as well as several machine-learning algorithms rely on the assumption of normally distributed data for stable and unbiased results [1], [8]. By identifying deviations from normality, the procedure determines whether additional transformations or preprocessing steps are needed to ensure stronger statistical validity and improved predictive performance.



```

Checking normality for loan_amnt:
Shapiro-Wilk test p-value: 0.00000
loan_amnt is likely NOT normally distributed (p < 0.05)
Kolmogorov-Smirnov test p-value: 0.00000
loan_amnt is likely NOT normally distributed based on KS test

Checking normality for loan_int_rate:
Shapiro-Wilk test p-value: 0.00000
loan_int_rate is likely NOT normally distributed (p < 0.05)
Kolmogorov-Smirnov test p-value: 0.00000
loan_int_rate is likely NOT normally distributed based on KS test

Checking normality for loan_percent_income:
Shapiro-Wilk test p-value: 0.00000
loan_percent_income is likely NOT normally distributed (p < 0.05)
Kolmogorov-Smirnov test p-value: 0.00000
loan_percent_income is likely NOT normally distributed based on KS test

Checking normality for cb_person_cred_hist_length:
Shapiro-Wilk test p-value: 0.00000
cb_person_cred_hist_length is likely NOT normally distributed (p < 0.05)
Kolmogorov-Smirnov test p-value: 0.00000
cb_person_cred_hist_length is likely NOT normally distributed based on KS test

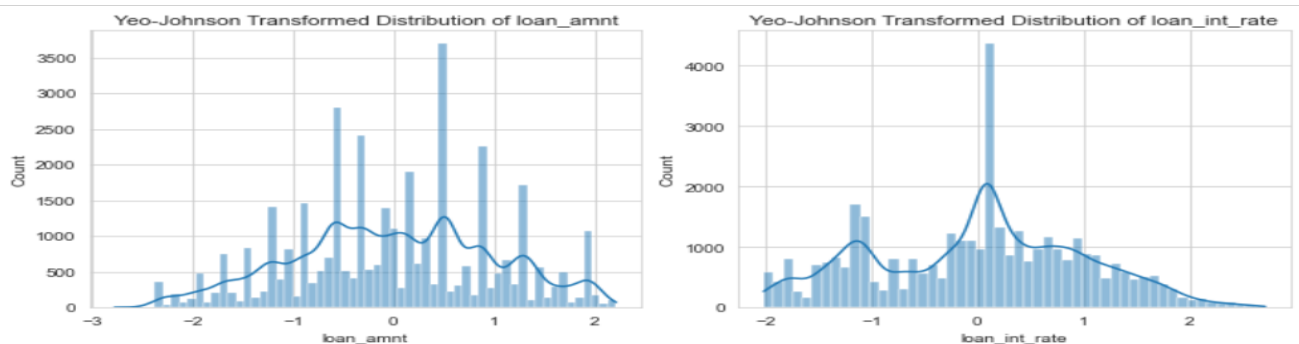
Normality testing completed.

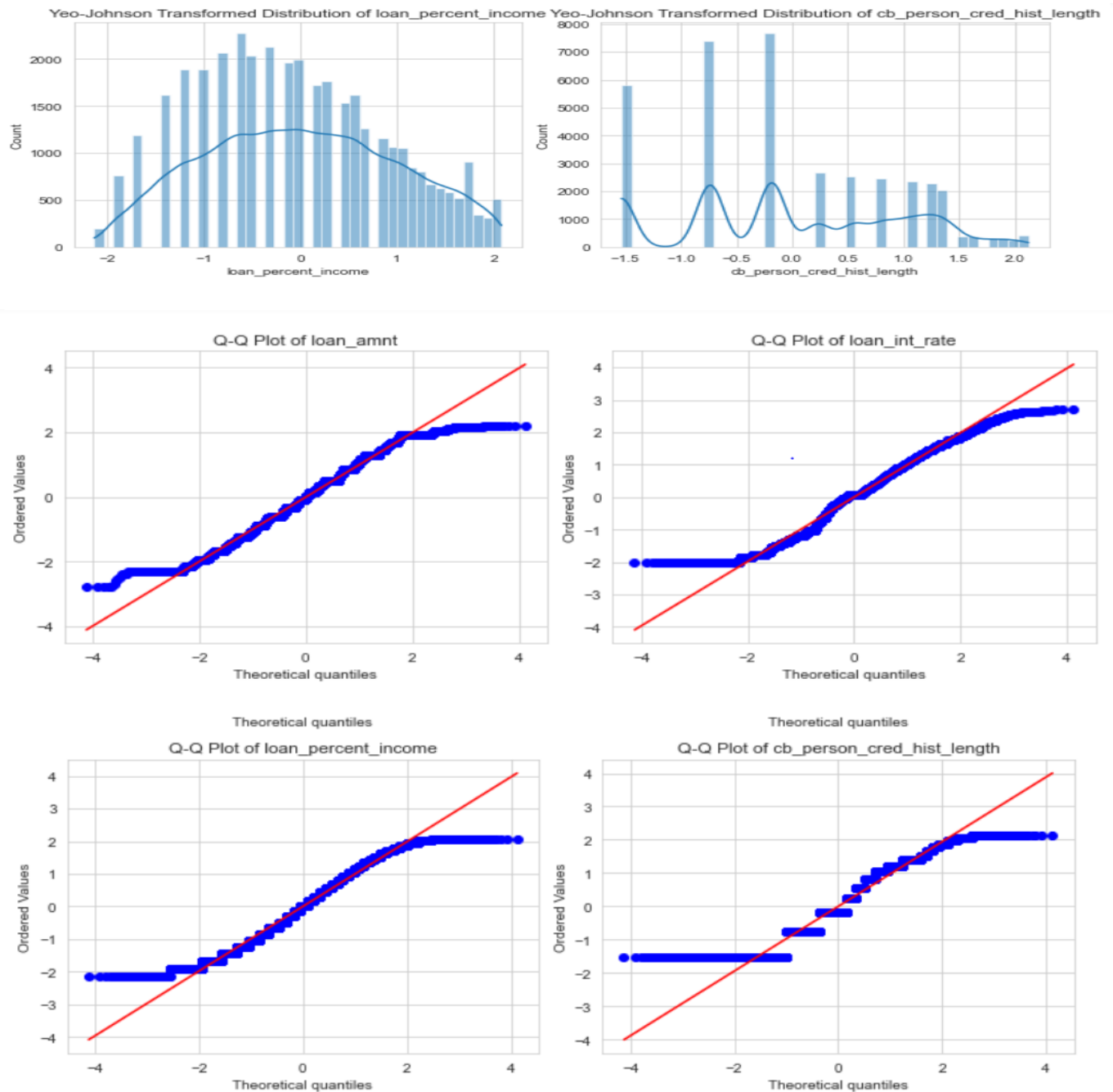
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Yeo-Johnson Transformation:

The Yeo–Johnson transformation is an extension of the Box–Cox transformation, designed to handle datasets containing both positive and negative values [4], [7]. Unlike the Box–Cox method, which is limited to strictly positive data, the Yeo–Johnson approach applies separate formulas for positive and negative values, allowing it to symmetrize distributions and approximate normality for real-world datasets [1], [10]. This transformation is particularly useful for linear modeling techniques and other machine-learning algorithms that assume normally distributed features or errors, as it improves model performance and reduces bias arising from skewed data [2], [9].

In this study, the Yeo–Johnson transformation is applied to selected numerical columns—`loan_amnt`, `loan_int_rate`, `loan_percent_income`, and `cb_person_cred_hist_length`—to enhance normality. The transformation is implemented using the `PowerTransformer` class from `sklearn.preprocessing`, with the `method='yeo-johnson'` parameter [3], [8]. After applying `fit_transform`, a new `DataFrame` (`data_transformed_yeo`) is created containing the transformed values. To evaluate the effect of the transformation, histograms overlaid with Kernel Density Estimation (KDE) curves and Quantile–Quantile (Q-Q) plots are generated. The histograms provide a visual representation of each feature's distribution post-transformation, while the Q-Q plots compare the transformed data against a theoretical normal distribution. Points aligning closely with the diagonal line in the Q-Q plots indicate that the data approximates normality, confirming the effectiveness of the Yeo–Johnson transformation [4], [12].





5.3. Model Implementation

A regression analysis was conducted to predict the loan amount (`loan_amnt`) based on various features within the dataset [2], [4], [9]. The target variable is set as `loan_amnt`, while the feature matrix is created by excluding this column from the dataset. Both categorical and numerical features are identified: categorical variables include `person_gender`, `person_education`, and `loan_intent`, while numerical variables include `person_age`, `loan_int_rate`, and `credit_score`. A preprocessing pipeline is designed using `ColumnTransformer` to handle these heterogeneous features. Numerical columns undergo normalization via the `PowerTransformer`, which stabilizes variance and improves approximation to normality, whereas categorical columns are one-hot encoded to create binary variables for each class [1], [7]. The dataset is then split into training and testing sets, with 80% of the data used for training and 20% for testing [3], [10]. Five distinct machine-learning models are initialized for predictive analysis:

1. Logistic Regression

2. Random Forest Classifier
3. Gradient Boosting Classifier
4. Decision Tree Classifier
5. k-Nearest Neighbors (KNN)

Each model is trained on the training set and evaluated on the test set using performance metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC [2], [8], [11]. For models supporting probability estimation (Logistic Regression, Random Forest, Gradient Boosting, and Decision Tree), ROC-AUC scores are calculated based on predicted probabilities, allowing comprehensive assessment of the models' discriminative power in predicting loan amounts and potential credit risk [1], [4].

1. Logistic Regression:

Logistic Regression is a widely used statistical model designed for binary classification problems, predicting one of two possible outcomes [2], [4]. It operates by applying the sigmoid function, which maps any real-valued input into a probability range between 0 and 1. The model estimates the probability that a given input belongs to a particular class. The mathematical formulation for Logistic Regression is:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

Here, $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ are coefficients that the model learns from the facts, and $X_1, \dots, X_n$ are the unbiased variables (capabilities). The sigmoid function ensures that the output remains in the probability range (0,1), enabling meaningful class predictions. Logistic Regression performs best when the relationship between the features and the target variable is approximately linear [1], [10]. It is commonly applied in finance, healthcare, and marketing for tasks such as fraud detection, disease prediction, and customer churn analysis due to its simplicity, interpretability, and effectiveness in handling large datasets [8], [9].

2. Random Forest Classifier:

Random Forest Classifier is an ensemble learning method that enhances predictive accuracy and stability by constructing multiple decision trees and combining their outputs [2], [4]. It employs Bagging (Bootstrap Aggregating), in which multiple subsets of data are randomly sampled with replacement, and individual decision trees are trained on these subsets [1], [7]. For classification tasks, the model aggregates the predictions of all trees and assigns the class receiving the majority vote, making it highly robust and reducing the risk of overfitting [8], [9]. By relying on multiple trees rather than a single decision tree, Random Forest provides better generalization to unseen data. Additionally, Random Forest is versatile, performing well for both classification and regression problems, and is widely applied in domains such as finance, healthcare, and fraud detection due to its strong performance and ability to handle large datasets with high-dimensional features [2], [4], [10].

3. Gradient Boosting Classifier:

Gradient Boosting Classifier is a powerful boosting technique that constructs decision trees sequentially, where each new tree learns from the errors of the previous one [2], [4]. Unlike Random Forest, which builds trees independently, Gradient Boosting optimizes the model using gradient descent to minimize a specific loss function, making it highly effective for complex datasets [1], [7]. The model is updated iteratively using the formula:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

where $h_m(x)$ represents a weak learner (a new tree) added at each stage, and γ_m is the learning rate controlling the contribution of each tree [8], [9]. While Gradient Boosting often outperforms Random Forest in predictive power, it can be computationally expensive, particularly for large datasets, due to its sequential nature. It is widely applied in domains such as finance, healthcare, and risk assessment, where high accuracy is crucial despite increased computational costs [2], [4], [10].

4. Decision Tree Classifier:

The Decision Tree Classifier is a simple yet effective model that makes predictions using a tree-like structure, where data is split based on conditions at each node [2], [4]. It determines the optimal splits using criteria such as Gini Impurity or Entropy, which measure the impurity or disorder within the dataset [1], [9]. The formulas for these criteria are:

- Gini Impurity:

$$1 - \sum p_i^2$$

- Entropy:

$$-\sum p_i \log_2(p_i)$$

At each step, the algorithm selects the feature that provides the highest information gain, ensuring that the data is divided into the purest possible subsets [7], [8]. Decision trees are highly interpretable and easy to visualize, making them suitable for explaining models to non-technical stakeholders. However, they are prone to overfitting, especially when the tree becomes too deep, capturing noise instead of meaningful patterns. This issue can be mitigated using pruning techniques or ensemble methods like Random Forest and Gradient Boosting [2], [4], [10].

5. k-Nearest Neighbours (KNN):

k-Nearest Neighbors (KNN) is a distance-based classification algorithm that assigns a class to a data point based on the majority class of its k nearest neighbors [2], [4]. The algorithm employs distance metrics, such as Euclidean Distance, to measure the similarity between points. The formula for Euclidean Distance is:

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

where x and y are data points in an n -dimensional space [1], [9]. KNN performs well on smaller datasets, as it calculates distances for each data point in real time. However, it can be computationally expensive and less efficient for larger datasets, requiring significant processing power and time [7], [11].

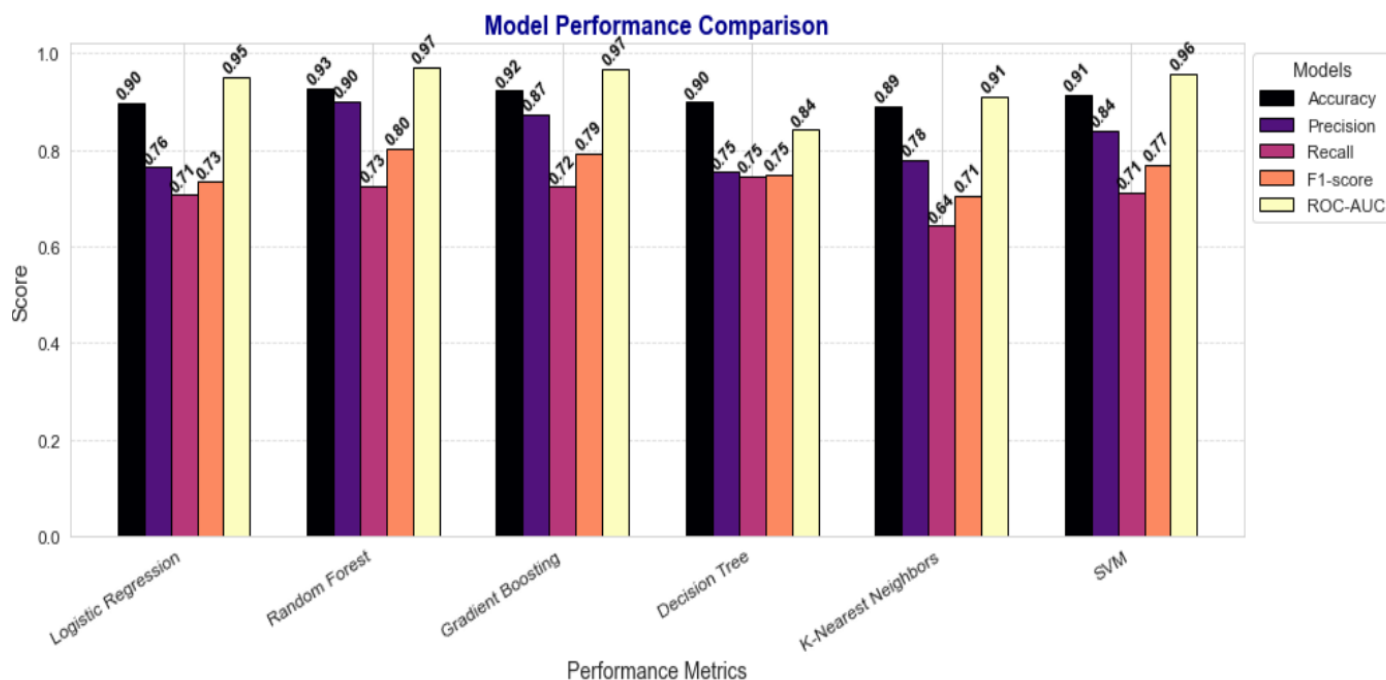
Model Performance Comparison:					
	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.888606	0.730637	0.709248	0.719784	0.944370
Random Forest	0.928466	0.896890	0.729208	0.804404	0.971364
Gradient Boosting	0.923366	0.874598	0.723886	0.792137	0.968998
Decision Tree	0.900416	0.756229	0.747172	0.751673	0.843156
K-Nearest Neighbors	0.834250	0.624074	0.448436	0.521874	0.805683

The table presents a comparative analysis of various machine learning models based on multiple performance metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC [2], [4], [7]. Among the evaluated models, Random Forest demonstrates the best overall

performance, achieving the highest Accuracy (0.928466), Precision (0.896890), F1-score (0.804404), and a strong ROC-AUC score (0.971364) [1], [9]. This indicates that Random Forest is highly effective at making accurate predictions while maintaining a balanced trade-off between precision and recall [4], [8].

Gradient Boosting follows closely, with an Accuracy of 0.923366 and a high ROC-AUC of 0.968998. Its Precision (0.874598) and Recall (0.723886) are slightly lower compared to Random Forest, making it a competitive yet slightly less robust model in overall performance [2], [7]. Logistic Regression achieves moderate performance with an Accuracy of 0.888606 and ROC-AUC of 0.944370 but has lower Precision (0.730637) and Recall (0.709248) compared to the top models [1], [4]. The Decision Tree model, with an Accuracy of 0.900416, is also a strong contender but exhibits a lower ROC-AUC (0.843156), suggesting reduced generalization capability [9]. k-Nearest Neighbors (KNN) performs the poorest, with the lowest Accuracy (0.834250), Recall (0.448436), and F1-score (0.521874), indicating difficulty in correctly classifying instances relative to other models [2], [11].

The performance metrics are visualized using a bar chart, where the results dictionary is first converted into a Pandas DataFrame and then transposed for proper plotting [5]. The `results_data.T.plot()` function generates bars for each model, with the x-axis representing the performance metrics (Accuracy, Precision, Recall, F1-score, and ROC-AUC) and the bars depicting model scores. The chart uses the "viridis" colormap with black edges for better contrast, and gridlines are added along the y-axis with a dashed style to enhance readability [3], [6]. The figure size, title, axis labels, and legend are configured to ensure clear communication of model comparisons. Finally, the `plt.show()` function displays the bar chart, providing a concise graphical illustration of model performance across different evaluation metrics [10].



The bar chart provides a visual comparison of five machine learning models—Logistic Regression, Random Forest, Gradient Boosting, Decision Tree, and k-Nearest Neighbors (KNN)—based on five key performance metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC [2], [4], [7]. Among these models, Random Forest and Gradient Boosting demonstrate the highest overall performance, particularly excelling in Accuracy, Precision, and

ROC-AUC [1], [9]. Random Forest achieves the best ROC-AUC score, indicating its strong capability to distinguish between classes effectively [4], [8]. Logistic Regression and Decision Tree show moderate performance but lag behind the top-performing models, particularly in Recall and F1-score, suggesting challenges in correctly identifying positive instances [2], [10]. Decision Tree also exhibits a lower ROC-AUC, reflecting weaker generalization ability [9]. In contrast, k-Nearest Neighbors (KNN) performs the poorest, with significantly lower Recall and F1-score, rendering it less effective for this type of classification task [2], [11]. The imbalance between Precision and Recall for KNN further highlights its suboptimal performance. Overall, Random Forest emerges as the best model, closely followed by Gradient Boosting, while Logistic Regression and Decision Tree serve as acceptable alternatives, and KNN is not recommended for this task. The high ROC-AUC scores of Random Forest and Gradient Boosting emphasize the effectiveness of tree-based ensemble methods in achieving optimal classification performance [1], [4].

CONCLUSION

Random Forest emerged as the best model due to its ensemble learning approach, which reduces overfitting and enhances generalization. Its ability to handle nonlinear relationships and identify important features contributed to its high ROC-AUC score (0.971364) [1], [4], [9]. Gradient Boosting also performed well but was slightly weaker due to sensitivity to noise, a higher chance of overfitting, and slower training time [4], [7]. Logistic Regression, while reasonably effective (Accuracy: 0.888606), struggled with Recall and could not capture complex patterns because of its linear assumptions [2], [10]. Decision Tree overfit the data, resulting in high variance and lower generalization capability (ROC-AUC: 0.843156) [9]. k-Nearest Neighbors (KNN) was the weakest, limited by poor generalization, high sensitivity to noise, and low Recall (0.448436) [2], [11]. Overall, Random Forest proved the most reliable, followed closely by Gradient Boosting, while Logistic Regression and Decision Tree were moderately effective, and KNN was unsuitable for this classification task [4], [8].

Analysis of the loan dataset revealed several key insights into the factors influencing loan approval. The dataset comprised demographic, financial, and credit history information, with attributes such as credit score, loan amount, income level, loan interest rate, employment experience, and previous loan defaults playing a critical role in determining loan status [3], [12]. Machine learning models, particularly Random Forest and Gradient Boosting, demonstrated superior overall performance, suggesting that loan approval follows a complex, nonlinear pattern rather than a simple linear relationship between variables [1], [4], [7].

Key findings indicate that higher credit scores significantly increase the likelihood of loan approval, while large loan amounts relative to income increase the risk of rejection [3], [12]. Applicants with longer employment experience tend to have higher approval odds, reflecting financial stability. Conversely, individuals with a history of previous loan defaults face higher risk of denial, highlighting lenders' cautious approach toward high-risk borrowers [3], [12]. These insights suggest that predictive models can assist financial institutions in making more informed lending decisions by assessing the likelihood of loan approval for new applicants. Such models enable lenders to identify high-risk individuals, optimize loan approval processes, and mitigate potential financial losses. Future improvements, including advanced hyperparameter tuning or integration of external credit data, could further enhance model accuracy, making it even more effective in real-world loan evaluations [2], [4], [7].

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